NLU Course Project 1: LM

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1. Introduction

This report illustrate my implementation and results for the first part of the NLU course project about improving the performance of language models for next-word prediction. It consists in two parts.

- **Part A:** Increase the performance of the baseline RNN by implementing the following techniques on top of each other: (i) Replacing the Recurrent Neural Network (RNN) with a Long-Short Term Memory (LSTM) network, (ii) Adding dropout layers (iii) Replacing Stochastic Gradient Descent (SGD) with AdamW.
- **Part B:** Starting from the best performing LSTM model from Part A apply the following regularization techniques: (i) Weight Tying, (ii) Variational Dropout, (iii) Non-monotonically Triggered AvSGD (NT-AvSGD)

2. Implementation details

Prior to each optimization technique, hyperparameters (hidden size, embedding size, dropout, and learning rate) were tuned to minimize perplexity, while training settings were fixed (100 epochs, patience 5, gradient clipping 5, batch size 64 training /128 evaluation). This tuning was performed via a 20-trial Optuna sweep [1], using a Tree-structured Parzen Estimator (TPE) to sample configurations by biasing the search toward regions that previously yielded lower validation loss. After all trials, the configuration with the lowest validation loss was selected as the final set of hyperparameters.

Part A has been implemented by:

1. Replacing the RNN with LSTM [2].
2. Adding two dropout layers, one after the embedding layer and one before the last linear layer [3], each with an independently tuned dropout probability.
3. Changing the optimizer from SGD to AdamW[4]

Part B has been implemented by increasingly applying the techniques discussed in this paper by Merity et al. [5]:

1. Implementing Weight Tying by sharing the weights between the embedding and softmax layers, which substantially reduces the total parameter count and enforces a common representation space [5]. Since weight tying requires the embedding and output dimensions to match, a linear projection layer was introduced whenever the sizes differed.
2. Applying Variational Dropout, a technique wherein a single binary dropout mask is sampled and reused across all time steps of a sequence within a given forward and backward pass. This approach ensures diversity in the dropped elements while maintaining temporal consistency to effectively regularize the model.
3. Implementing NT-AvSGD, a variant of AvSGD in which averaging is triggered when the validation metric fails to improve for multiple cycles. Specifically, averaging begins when the current validation loss exceeds the minimum value recorded within a sliding window defined by a non-monotone interval. From that point onward, model parameters are averaged until training ends, leading to more stable optimization and better generalization. This technique was adapted in Python based on the pseudocode provided in [5].

3. Results

The results successfully met the target of achieving a perplexity below 250. Interestingly, the best-performing models across the different parts did not share the same embedding and hidden sizes, suggesting that performance depends more on finding the right balance between model capacity and the optimization technique employed than on simply increasing model dimensions. The following sections provide insights and conclusion to each part.

Part A: All steps progressively reduced perplexity, resuls are shown in Table 1.

- **Baseline LSTM:** Replacing the vanilla RNN with an LSTM reduced perplexity from 153.11 to 137.59, confirming the better capturing of long-term relations and mitigatin of the vanishing gradient.
- **Dropout:** Introducing dropout substantially improved generalization, reducing perplexity to 113.07.
- **AdamW:** Replacing SGD with AdamW yielded the best result in Part A, achieving a perplexity of 107.59.The optimizer converged effectively with a much smaller learning rate, demonstrating the benefits of adaptive learning rates and decoupled weight decay.

Part B: Weight Tying, Variational Dropout, and NT-AvSGD were evaluated on top of the baseline LSTM. Results are reported in Table 2.

- **Weight Tying:** Reduced perplexity to 117.21. This suggests that sharing input and output word representations can improve parameter efficiency without sacrificing performance.
- **Variational Dropout:** Further decreased perplexity to 98.71. Applying a fixed dropout mask across time steps likely provided stronger regularization while preserving temporal consistency within sequences.
- **NT-AvSGD:** NT-AvSGD achieved the best overall result, reaching a perplexity of 87.70. Averaging model parameters after convergence appears to have improved optimization stability and helped the model converge to a better-performing solution.

4. References

[1] T. Akiba, S. Sano, T. Yanase, T. Ohta, and M. Koyama, "Optuna: A next-generation hyperparameter optimization framework," in *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*. ACM, 2019, pp. 2623–2631. [Online]. Available: https://doi.org/10.1145/3292500.3330701

[2] PyTorch Developers, *torch.nn.LSTM — PyTorch Documentation*, PyTorch, 2025, accessed: 2026-01-27. [Online]. Available: https://docs.pytorch.org/docs/stable/generated/torch.nn.LSTM.html

[3] ——, *torch.nn.Dropout — PyTorch Documentation*, PyTorch, 2025, accessed: 2026-01-27. [Online]. Available: https://docs.pytorch.org/docs/stable/generated/torch.nn.Dropout.html

[4] ——, *torch.optim.AdamW — PyTorch Documentation*, PyTorch, 2025, accessed: 2026-01-27. [Online]. Available: https://docs.pytorch.org/docs/stable/generated/torch.optim.AdamW.html

[5] S. Merity, N. S. Keskar, and R. Socher, "Regularizing and optimizing lstm language models," *arXiv preprint arXiv:1708.02182*, 2017.

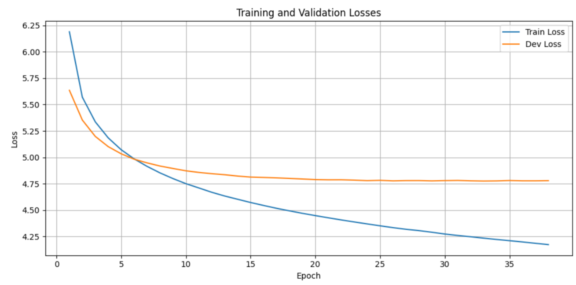
Model	Hid size	Emb size	Lr	Emb dropout	Out dropout	PPL (Test)
Baseline RNN	350	400	0.5	–	–	153.11
Baseline LSTM	400	450	3.0	–	–	137.59
+ Dropout	400	400	0.5	0.1	0.5	113.07
+ AdamW	350	450	0.001	0.4	0.5	**107.59**

Table 1: *Part A: Hyperparameter values and perplexity on test dataset for the best saved models. The best result is highlighted.*

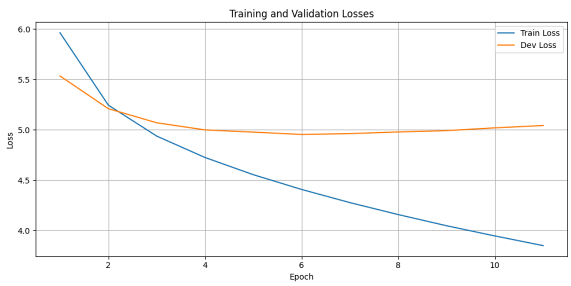
Model	Hid size	Emb size	Lr	Emb dropout	Out dropout	PPL (Test)
Baseline LSTM	400	450	3.0	–	–	137.59
+ Weight Tying	300	300	5.0	–	–	117.21
+ Variational Dropout	350	450	3.0	0.5	0.4	98.71
+ NT-AvSGD	400	300	2.0	0.5	0.5	**87.70**

Table 2: *Part B: Hyperparameter values and perplexity on test dataset for the best saved models. The best result is highlighted.*

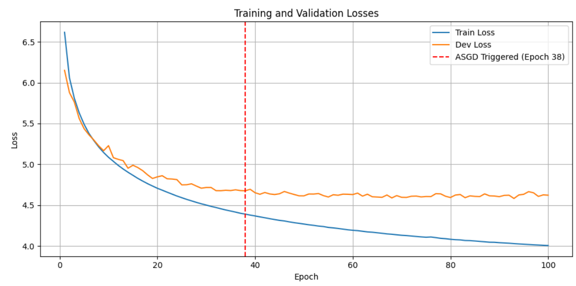
(a) *Training and validation losses for Baseline RNN.*



(b) *Training and validation losses for LSTM with Dropout layers and AdamW.*



(c) *Training and validation losses for baseline LSTM.*



(d) *Training and validation losses for LSTM with Weight Tying, Variational Dropout and NT-AvSGD.*

Figure 1: *Loss plots of the baseline and best performing models for both parts.*